Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

A. Epstein-Barr virus (EBV)

B. Measles virus

C. Human immunodeficiency virus (HIV)

D. Human papillomavirus

Reasoning + Answer:

<think>

- CD21 is a receptor on B cells that helps in the activation of B cells by the interaction with CD19 and CD81. It also plays a role in B cell signaling and antibody production.
- EBV and measles both have the ability to infect B cells and use them for their replication and spread. Specifically, EBV uses CD21 as a receptor to enter B cells, which are crucial for its replication cycle.
- HIV primarily targets and destroys CD4+ T cells, not B cells.
- Human papillomavirus (HPV) infects epithelial cells and does not have a dependency on CD21 for its infection cycle.

Given these points, the genetic modification of B cells not to express CD21 would most affect the pathogenesis of Epstein-Barr virus (EBV), as EBV uses CD21 as a receptor for infecting B cells.

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<answer>A. Epstein-Barr virus (EBV)</answer>